

Spearman’s Rank Correlation Coefficient

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Rank correlation

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In 1904, Charles Edward Spearman, a British psychologist found out the method of ascertaining the coefficient of correlation by ranks. This method is based on rank. This measure is useful in dealing with qualitative characteristics, such as intelligence, beauty, morality, character, etc. It cannot be measured quantitatively, as in the case of Pearson’s coefficient of correlation.

Rank correlation is applicable only to individual observations. The result we get from this method is only an approximate one, because under ranking method original value are not taken into account. The formula for Spearman’s rank correlation which is denoted by ρ (pronounced as row) is

$$\rho = 1 - \frac{6\sum d^2}{N(N^2 - 1)} \quad \text{or}$$

$$\rho = 1 - \frac{6\sum d^2}{N^3 - N}$$

where d = The difference of two ranks = $R_X - R_Y$ and

N = Number of paired observations.

Rank coefficient of correlation value lies between -1 and $+1$. Symbolically, $-1 \leq \rho \leq +1$

When we come across spearman’s rank correlation, we may find two types of problem

(i) When ranks are given

(ii) When ranks are not given

Example 9.6

The following are the ranks obtained by 10 students in Statistics and Mathematics

Statistics	1	2	3	4	5	6	7	8	9	10
Mathematics	1	4	2	5	3	9	7	10	6	8

Find the rank correlation coefficient.

Solution:

Let R_X is considered for the ranks of Statistics and R_Y is considered for the ranks of mathematics.

R_x	R_y	$d = R_X - R_Y$	d^2
1	1	0	0
2	4	-2	4
3	2	1	1
4	5	-1	1
5	3	2	4
6	9	-3	9
7	7	0	0
8	10	-2	4
9	6	3	9
10	8	2	4
			$\sum d^2 = 36$

Table 9.4

The rank correlation is given by

$$\rho = 1 - \frac{6\sum d^2}{N(N^2 - 1)} = 1 - \frac{6(36)}{10(10^2 - 1)}$$

$$= 1 - 0.218$$

$$\therefore \rho = 0.782$$

Example 9.7

Ten competitors in a beauty contest are ranked by three judges in the following order

First judge	1	4	6	3	2	9	7	8	10	5
Second judge	2	6	5	4	7	10	9	3	8	1
Third judge	3	7	4	5	10	8	9	2	6	1

Use the method of rank correlation coefficient to determine which pair of judges has the nearest approach to common taste in beauty?

Solution:

Let R_X, R_Y, R_Z denote the ranks by First judge, Second judge and third judge respectively

R_x	R_y	R_z	$d_{xy} = R_x - R_y$	$d_{yz} = R_y - R_z$	$d_{zx} = R_z - R_x$	d_{xy}^2	d_{yz}^2	d_{zx}^2
1	2	3	-1	-1	2	1	1	4
4	6	7	-2	-1	3	4	1	9
6	5	4	1	1	-2	1	1	4
3	4	5	-1	-1	2	1	1	4
2	7	10	-5	-3	8	25	9	64
9	10	8	-1	2	-1	1	4	1
7	9	9	-2	0	2	4	0	4
8	3	2	5	1	-6	25	1	36
10	8	6	2	2	-4	4	4	16
5	1	1	4	0	-4	16	0	16
						$\sum d_{xy}^2 = 82$	$\sum d_{yz}^2 = 22$	$\sum d_{zx}^2 = 158$

$$\rho_{XY} = 1 - \frac{6\sum d_{XY}^2}{N(N^2 - 1)} = 1 - \frac{6(82)}{10(10^2 - 1)} = 1 - 0.4969 = 0.5031$$

$$\rho_{YZ} = 1 - \frac{6\sum d_{YZ}^2}{N(N^2 - 1)} = 1 - \frac{6(22)}{10(10^2 - 1)} = 1 - \frac{132}{990}$$

$$= 1 - 0.1333 = 0.8667$$

$$\rho_{ZX} = 1 - \frac{6\sum d_{ZX}^2}{N(N^2 - 1)} = 1 - \frac{6(158)}{10(10^2 - 1)}$$

$$= 1 - 0.9576 = 0.0424$$

Since the rank correlation coefficient between Second and Third judges i.e., ρ_{YZ} is positive and weight among the three coefficients. So, Second judge and Third judge have the nearest approach for common taste in beauty.

Example 9.8

Calculate rank correlation coefficient of the following data

Subject 1	40	46	54	60	70	80	82	85	87	90	95
Subject2	45	46	50	43	40	75	55	72	65	42	70

Solution:

Let X is considered for Subject1 and Y is considered for Subject2

X	Y	R_x	R_y	$d = R_X - R_Y$	d^2
40	45	1	4	-3	9
46	46	2	5	-3	9
54	50	3	6	-3	9
60	43	4	3	1	1
70	40	5	1	4	16
80	75	6	11	-5	25
82	55	7	7	0	0
85	72	8	10	-2	4
87	65	9	8	1	1
90	42	10	2	8	64
95	70	11	9	2	4
					$\sum d^2 = 142$

Table 9.6

$$\rho = 1 - \frac{6\sum d^2}{N(N^2 - 1)}$$

$$\rho = 1 - \frac{6(142)}{11(11^2 - 1)}$$

$$\rho = 1 - \frac{852}{1320} = 0.354$$

Exercise 9.1

1. Calculate the correlation co-efficient for the following data

X	5	10	5	11	12	4	3	2	7	1
Y	1	6	2	8	5	1	4	6	5	2

2. Find coefficient of correlation for the following:

Cost (Rs.)	14	19	24	21	26	22	15	20	19
Sales (Rs.)	31	36	48	37	50	45	33	41	39

3. Calculate coefficient of correlation for the ages of husbands and their respective wives:

Age of husbands	23	27	28	29	30	31	33	35	36	39
Age of wives	18	22	23	24	25	26	28	29	30	32

4. Calculate the coefficient of correlation between X and Y series from the following data

	X	Y
Number of pairs of observation	15	15
Arithmetic mean	25	18
Standard deviation	3.01	3.03
Sum of squares of deviation from the arithmetic mean	136	138

Summation of product deviations of X and Y series from their respective arithmetic means is 122.

5. Calculate correlation coefficient for the following data

X	25	18	21	24	27	30	36	39	42	48
Y	26	35	48	28	20	36	25	40	43	39

6. Find coefficient of correlation for the following:

X	78	89	96	69	59	79	68	62
Y	121	72	88	60	81	87	123	92

7. An examination of 11 applicants for a accountant post was taken by a finance company. The marks obtained by the applicants in the reasoning and aptitude tests are given below.

Applicant	A	B	C	D	E	F	G	H	I	J	K
Reasoning test	20	50	28	25	70	90	76	45	30	19	26
Aptitude test	30	60	50	40	85	90	56	82	42	31	49

Calculate Spearman’s rank correlation coefficient from the data given above.

8. The following are the ranks obtained by 10 students in commerce and accountancy are given below

Commerce	6	4	3	1	2	7	9	8	10	5
Accountancy	4	1	6	7	5	8	10	9	3	2

To what extent is the knowledge of students in the two subjects related?

9. A random sample of recent repair jobs was selected and estimated cost and actual cost were recorded.

Estimated cost	300	450	800	250	500	975	475	400
Actual cost	273	486	734	297	631	872	396	457

Calculate the value of spearman’s correlation coefficient.

10. The rank of 10 students of same batch in two subjects A and B are given below. Calculate the rank correlation coefficient.

Rank of A	1	2	3	4	5	6	7	8	9	10
Rank of B	6	7	5	10	3	9	4	1	8	2